

Dyadic Scaling Rigidity and Log-Periodic Growth of 3AP-Free Permutations

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Abstract

Let $\theta(n)$ count permutations of $[n]$ having no three-term arithmetic progression as a subsequence. A single exponential growth constant does not exist, by a theorem of Boon Suan Ho. We determine a more informative asymptotic object: a unique continuous function P , periodic in the binary logarithm, with

$$\log \theta(n) = nP(\log_2 n) - Q(n), \quad \log 2 \leq Q(n) \leq \log 21.$$

The construction is an application of established divide-and-conquer theory; an elementary proof with explicit constants is supplied. The central further result is an exact scale-rigidity theorem. Published enumerative data certify that P has fundamental period exactly one. Consequently, the positive real scales c for which $\log \theta(\lfloor cn \rfloor) - c \log \theta(n) = o(n)$ are precisely the integral powers of two. At every other scale, exponential gain and exponential loss both occur on sets of positive lower natural density. For the concrete split $3n = 2n + n$, integer certificates give a factor $(49/50)^n$ loss when $n = 128 \cdot 2^t$, and a factor $(21/20)^n$ gain when $n = 172 \cdot 2^t$. We also describe the entire interval of subsequential growth rates, all cutoff-dependent empirical limit laws, logarithmic averaging, and polynomial and geometric sampling. A finite-data enclosure algorithm, reproducible arithmetic certificates, an independent small-count verification, and a research program complete the article.

Research status. Unrefereed mathematical development, not a Lean-verified result. The general periodic-profile method and the failure of a single growth constant are credited to prior work. The scale-rigidity and statistical consequences are derived here; their priority in the entire literature has not been certified. Numerical certificates use published large counts, independently recomputed in this package only through $n = 32$.

Collection note [write]. This article is the report `a003407-dyadic-scaling-rigidity` of PROVEIT's research-report collection, written from one manuscript (batch 40, manuscript 06; see Section 14). None of its own theorems is formalized. Two inputs that the manuscript takes from the literature, Sharma's upper splitting bound and the parity lower recurrence, are already proved in Lean in PROVEIT; Section 11 names the declarations and says what they do and do not cover. Placement of this report near a Lean development gives it no formal status. Text added when the report was written is marked [write].

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1 The question and the contribution

1.1 What should replace a nonexistent growth constant?

Many counting problems are organized around an asymptotic expression of the form $C\rho^n n^\alpha$. For three-term-progression-free permutations, even the coarsest candidate $\rho = \lim \theta(n)^{1/n}$ fails to exist. Merely recording this failure leaves several natural structural questions unanswered. Does the sequence exhibit arbitrary exponential-scale oscillation, or a coherent phase law? Which changes of scale preserve its growth? Can differently sized blocks be combined at only subexponential cost? Does averaging eliminate the oscillation?

This article answers these questions in a unified way. The right object is a continuous growth profile on the circle $\mathbb{R}/\mathbb{Z}$, sampled at the phase $\log_2 n$. Its period is not just some divisor of one: a short finite certificate proves that its fundamental period is exactly one. This numerical shape information has a qualitative consequence for *every real scale*, not merely the finitely many sizes in the table.

Here is the central classification. In this display, $c > 0$ is fixed and the integer $\lfloor cn \rfloor$ is positive for all sufficiently large n :

$$\log \theta(\lfloor cn \rfloor) - c \log \theta(n) = o(n) \iff c \in \{2^j : j \in \mathbb{Z}\}. \quad (1.1)$$

Outside these scales, the left side has positive and negative excursions of order n , each occurring on a set of positive lower natural density. Thus the obstruction is substantially stronger than failure of a ratio limit.

1.2 Relationship to existing work

Davis, Entringer, Graham, and Simmons established the parity-splitting construction for these permutations [1]. Sharma proved the uniform upper splitting estimate used below [2, Theorem 2.8]. Correll and Randy W. Ho developed dynamic programming and extended the count table [4]; the OEIS table now supplies values through 200, with later terms attributed to Alois P. Heinz [7].

Boon Suan Ho proved in 2026 that $\theta(n)^{1/n}$ does not converge, by separating limits along two dyadic subsequences [5]. That theorem is *not* claimed as a result of this article. Separately, Hwang, Janson, and Tsai give a general theory of continuous periodic fluctuations for floor/ceiling divide-and-conquer recurrences [6, Theorem 2.10]. Our profile construction is the equal-weight, bounded-toll specialization of that theory. We give the proof because the explicit bounds and interpolation are needed for the subsequent certificates, not because the underlying analytic principle is new.

The additional contribution developed here consists of the exact period certificate, the resulting classification (1.1), exponential obstructions to unequal-size gluing, and the complete sampling and empirical laws for this particular counting sequence. These statements were not found in the sources inspected for this article. This is a targeted literature assessment, not a claim of exhaustive priority verification or of resolution of a separately named published conjecture.

1.3 Connection with the repository

The selected topic belongs directly to the Ramsey-theoretic part of PROVEIT [8]. In particular, the repository contains the directories

Combinatorics/Ramsey/Lean/DavisEntringerGrahamSimmons1977/

Combinatorics/Ramsey/Lean/LeSaulnierVijay2011/.

The latter statement catalogue, based on [3], uses $M(n)$ for the count denoted here by $\theta(n)$, records the even and odd parity recurrences, and includes the exponential bounds from the literature. A proposition-valued declaration in a statement catalogue is not by itself a proved theorem. We therefore take the published combinatorial estimates as the mathematical input, rather than inferring proof status from a declaration name.

The inspected repository revision was

74f7f5bdba1aa728b340509701a0fa4e45e8dbf3.

No repository files were changed, and no full Lean build or axiom audit was performed. The article extends the research direction, not the repository’s machine-checked theorem inventory.

Remark 1.1 (Correction [write]). The caution above is sound as method, but its conclusion is stale. At the inspected revision, and unchanged at the time of writing, the declarations in question are not bare statements. The two parity recurrences are proved as `LeanProofs.LeSaulnierVijay2011.M_even_recurrence_holds` and `M_odd_recurrence_holds`, which are derived from the Davis–Entringer–Graham–Simmons declarations `even_count_recurrence_holds` and `odd_count_recurrence_holds`. Sharma’s Theorem 2.8, which this article imports, is proved as `LeanProofs.Sharma2012.theorem_2_8_holds`. The manuscript read only the statement catalogues, so it did not see these proofs. Section 11 gives the exact names, files and scope. The mathematics below is unaffected: both inputs are true and are now also machine-checked in PROVEIT. The article’s own results remain unformalized.

1.4 Dependency and verification boundary

The one substantial theorem imported without reproducing its complete combinatorial proof is Sharma’s upper recurrence. All subsequent analytic and structural deductions are proved below. The exact-period and explicit-gluing certificates also use the published count table. The supplied program verifies the required integer inequalities, but that does not independently prove the large counts themselves. An independent subset-state enumeration was executed for every $0 \leq n \leq 32$, and direct permutation enumeration for $0 \leq n \leq 8$. These are separate checks with separate evidential roles.

[write] Sharma’s recurrence is imported by this article, but it is not an unchecked input in PROVEIT: its Lean proof is named in Section 11. The published count table beyond $n = 20$ has no counterpart in PROVEIT; the Lean development certifies $\theta(1), \dots, \theta(20)$ only, whereas the numerical enclosures of Section 4 and the certificates of Sections 5–6 use published counts with $100 \leq n \leq 200$.

1.5 Notation, and the names used in the repository [write]

No symbol of the manuscript has been renamed. The table records the letters that carry more than one meaning here, or a different meaning in the Lean developments of PROVEIT, with the tempting false reading beside the true one.

Symbol	Meaning in this article	Remark; tempting false reading
$\theta(n)$	number of 3AP-free permutations of $[n]$ (OEIS A003407)	equals Lean's <code>Sharma2012.theta n</code> and the <code>M n</code> of the Davis–Entringer–Graham–Simmons and LeSaulnier–Vijay developments (<code>Sharma2012.theta_eq_davis_count</code>)
$M(x)$	cutoff-dependent limit of a Cesàro mean, $1 \leq x \leq 2$ (Theorem 7.3 only)	not the count $M(n) = \theta(n)$ of the Lean developments and of [1, 3]
$R(x)$	piecewise-affine interpolation of the toll r_n , (3.2)	not the certificate ratios $R_{\pm}$ of Theorem 6.4
$C, C_f, C(S)$	$\int_1^2 \Phi$; $\int_1^2 f \circ \Phi$; the subset-state continuation count (10.2)	three different objects
Q	bounded correction in (3.4), values in $[\log 2, \log 21]$	not the toll r_n (whose interpolation is R)
L, U, D	the constants $\log 2, \log 21, \log(21/2)$	not $\liminf b_n$ and $\limsup b_n$ (those are $p_{\pm}$)
γ	$\log_2(21/2)$ in (9.3)	an exponent, not a growth rate

Unqualified logarithms are natural, and $\log_2$ is the binary logarithm.

2 Combinatorial input and the bounded toll

Definition 2.1. A permutation $\pi = (\pi_1, \dots, \pi_n)$ of $[n] = \{1, \dots, n\}$ is *3AP-free* if there are no indices $i < j < k$ such that $\pi_i + \pi_k = 2\pi_j$. Let $\theta(n)$ be the number of these permutations, with $\theta(0) = 1$. Both increasing and decreasing arithmetic progressions are forbidden; a progression need not occupy consecutive positions.

For example, $\theta(1) = 1$, $\theta(2) = 2$, $\theta(3) = 4$, and $\theta(4) = 10$. The distinction between consecutive positions and arbitrary subsequences is essential to the count.

Lemma 2.2 (Parity splitting). *For every integer $n \geq 2$,*

$$\theta(n) \geq 2\theta(\lfloor n/2 \rfloor)\theta(\lceil n/2 \rceil). \quad (2.1)$$

Proof. Choose a 3AP-free ordering of the even numbers and one of the odd numbers in $[n]$, obtained by affine rescaling of smaller 3AP-free permutations. Concatenate the two blocks, in either order. The two endpoints of any integer three-term progression have the same parity. If its endpoints were in one block and its midpoint in the other, the midpoint could not lie between the endpoints in the concatenated order. If all three entries were in one block, that block's avoidance property would be violated. Thus both concatenations avoid three-term progressions. Their orders and their restrictions to the two nonempty parity blocks determine the input pair, so the construction counts exactly twice the number of input pairs. $\square$

Theorem 2.3 (Sharma's upper splitting bound). *For $n \geq 3$,*

$$\theta(n) \leq 21\theta(\lfloor n/2 \rfloor)\theta(\lceil n/2 \rceil). \quad (2.2)$$

The same inequality is valid at $n = 2$ by direct evaluation.

This is [2, Theorem 2.8]. The proof uses structural restrictions on how fixed odd and even traces can interleave. We do not replace that combinatorial theorem by a numerical fit or an unverified recurrence.

Remark 2.4 (Formal status of the two inputs [write]). Inequality (2.2) for $n \geq 3$ is the Lean theorem `LeanProofs.Sharma2012.theorem_2_8_holds`; the case $n = 2$ is the direct evaluation $\theta(2) = 2 \leq 21$, which the Lean statement does not include. Inequality (2.1) for all $n \geq 2$ is the pair of Lean theorems `even_count_recurrence_holds` ($n = 2k$) and `odd_count_recurrence_holds` ($n = 2k + 1$) in the namespace `LeanProofs.DavisEntringerGrahamSimmons1977`, for $k \geq 1$. The proof of the parity-splitting lemma above is the manuscript's own, and is a second route to a result already machine-checked. See Section 11.

For the remainder of the paper set

$$a_n = \log \theta(n), \quad b_n = \frac{a_n}{n}, \quad L = \log 2, \quad U = \log 21, \quad D = U - L = \log(21/2), \quad (2.3)$$

where unqualified logarithms are natural. Define the *toll*

$$r_n = a_n - a_{\lfloor n/2 \rfloor} - a_{\lceil n/2 \rceil}, \quad n \geq 2. \quad (2.4)$$

Equations (2.1)–(2.2) say precisely that

$$L \leq r_n \leq U. \quad (2.5)$$

This is an exact recurrence with a bounded, generally unknown toll; it is not an assertion that the toll converges or has any simple closed form.

Lemma 2.5 (Elementary global bounds). *For every $n \geq 1$,*

$$(n - 1)L \leq a_n \leq (n - 1)U. \quad (2.6)$$

Proof. The assertion holds at $n = 1$. If it holds at the two smaller arguments in (2.4), their sizes sum to n , and their lower bounds sum to $(n - 2)L$. Adding $r_n \geq L$ gives the lower bound at n ; the upper bound is identical with U in place of L . Strong induction completes the proof. $\square$

3 A continuous growth profile with a bounded correction

3.1 The interpolation identity

Let $A : [1, \infty) \rightarrow \mathbb{R}$ be the piecewise-affine interpolation of a_n :

$$A(n + u) = (1 - u)a_n + ua_{n+1}, \quad 0 \leq u \leq 1. \quad (3.1)$$

For $x \geq 2$, define

$$R(x) = A(x) - 2A(x/2). \quad (3.2)$$

Lemma 3.1 (No interpolation error). *R is the piecewise-affine interpolation of r_n . In particular,*

$$L \leq R(x) \leq U, \quad |R(x) - R(y)| \leq \min\{D|x - y|, D\} \quad (x, y \geq 2). \quad (3.3)$$

Proof. At even integers $n = 2m$ we have $2A(n/2) = 2a_m$; at odd integers $n = 2m + 1$ we have $2A(n/2) = a_m + a_{m+1}$. Thus $R(n) = r_n$ in both cases. On each interval $[n, n + 1]$, both $A(x)$ and $A(x/2)$ are affine: a breakpoint of the latter can only occur at an even integer. Consequently R is affine there, with endpoint values r_n, r_{n+1} . These values lie in $[L, U]$, so its slope has absolute value at most D , and its entire range has diameter at most D . $\square$

3.2 Construction and uniqueness

Theorem 3.2 (Bounded-error log-periodic law). *There is a unique continuous 1-periodic function $P : \mathbb{R} \rightarrow \mathbb{R}$ such that $a_n = nP(\log_2 n) + O(1)$. More precisely, there is a continuous function $Q : [1, \infty) \rightarrow [L, U]$ for which*

$$A(x) = xP(\log_2 x) - Q(x), \quad x \geq 1. \quad (3.4)$$

In particular,

$$\frac{e^{nP(\log_2 n)}}{21} \leq \theta(n) \leq \frac{e^{nP(\log_2 n)}}{2}, \quad n \geq 1. \quad (3.5)$$

Also $L \leq P(t) \leq U$ for every t .

Proof. For $x \geq 1$ and $k \geq 0$ put $F_k(x) = 2^{-k}A(2^k x)$. By (3.2),

$$F_{k+1}(x) - F_k(x) = 2^{-(k+1)}R(2^{k+1}x). \quad (3.6)$$

The uniform bound on R makes this a uniformly convergent telescoping series, indeed uniformly on all $x \geq 1$. Its limit is

$$F(x) = A(x) + \sum_{j=1}^{\infty} 2^{-j}R(2^j x). \quad (3.7)$$

Each summand is continuous, so F is continuous. The tail estimate is

$$L2^{-k} \leq F(x) - F_k(x) \leq U2^{-k}. \quad (3.8)$$

Shifting the defining limit by one index gives $F(2x) = 2F(x)$.

Define $P(t) = 2^{-t}F(2^t)$ initially for $t \geq 0$. The scaling identity gives $P(t+1) = P(t)$ and matching endpoint values, so this defines a continuous 1-periodic function on all of $\mathbb{R}$. Taking

$$Q(x) = F(x) - A(x) = \sum_{j \geq 1} 2^{-j}R(2^j x) \quad (3.9)$$

gives $L \leq Q(x) \leq U$ and proves (3.4). Exponentiation at integer $x = n$ gives (3.5). Interpolating (2.6) and taking the scaled limit gives $Lx \leq F(x) \leq Ux$, which proves the asserted bounds on P .

For uniqueness, suppose a second continuous 1-periodic function P_1 satisfies the same bounded-error assertion. Fix $t \in [0, 1]$ and take $n_k = \lfloor 2^{k+t} \rfloor$. Then $\log_2 n_k - k \rightarrow t$. Division of both asymptotic formulas by n_k and continuity imply $P(t) = \lim b_{n_k} = P_1(t)$. Periodicity extends this equality to every real argument. $\square$

The theorem is the equal-weight bounded-toll case of the framework in [6]; the proof above makes the constants needed here explicit. The bound on Q is a genuine uniform multiplicative control on $\theta(n)$, not just a relative $o(1)$ error in its logarithm. It does *not* assert that $Q(n)$ converges, so it should not be rewritten as an asymptotic equivalence with a fixed prefactor.

3.3 A quantitative modulus of continuity

Write

$$\Phi(s) = \frac{F(s)}{s} = P(\log_2 s), \quad 1 \leq s \leq 2. \quad (3.10)$$

We call s the binary mantissa. The endpoints are identified, since $\Phi(1) = \Phi(2)$.

Proposition 3.3 (Explicit log-Lipschitz control). *For $s, t \in [1, 2]$, $0 < h = |s - t| \leq 1/2$, and $J = \lceil \log_2(1/h) \rceil$,*

$$|\Phi(s) - \Phi(t)| \leq h[L + U + D(J + 1)]. \quad (3.11)$$

Consequently P is Hölder continuous of every exponent less than one, and has a circle modulus $O(h \log(1/h))$ as $h \downarrow 0$.

Proof. On $[1, 2]$, $A(s) = L(s - 1)$. In the series (3.7), each of the first J terms changes by at most Dh when s is replaced by t , by the Lipschitz bound in (3.3). The remaining terms change by at most $D \sum_{j>J} 2^{-j} = D2^{-J} \leq Dh$. Hence $|F(s) - F(t)| \leq Lh + Dh(J + 1)$. Finally,

$$\left| \frac{F(s)}{s} - \frac{F(t)}{t} \right| \leq |F(s) - F(t)| + Uh,$$

because $F(t) \leq Ut$ and $s, t \geq 1$. This proves (3.11). Composition with $s = 2^t$ transfers the estimate to each phase interval. Across the identified endpoints, split the distance at the endpoint and use $\Phi(1) = \Phi(2)$. The Hölder assertions follow from $h \log(1/h) = O_\alpha(h^\alpha)$ for $0 < \alpha < 1$. $\square$

We do not claim that this modulus is optimal for the particular sequence, or that P is differentiable, Lipschitz, or nowhere differentiable.

4 The growth spectrum and finite-data reconstruction

4.1 Every intermediate growth rate occurs

Define

$$p_- = \min_{t \in [0,1]} P(t), \quad p_+ = \max_{t \in [0,1]} P(t), \quad \rho_- = e^{p_-}, \quad \rho_+ = e^{p_+}. \quad (4.1)$$

Theorem 4.1 (Complete subsequential spectrum). *The sets of subsequential limits are*

$$\text{Clust}(b_n) = [p_-, p_+], \quad \text{Clust}(\theta(n)^{1/n}) = [\rho_-, \rho_+]. \quad (4.2)$$

For each $s \in [1, 2]$,

$$\frac{\log \theta(\lfloor 2^k s \rfloor)}{\lfloor 2^k s \rfloor} \longrightarrow \Phi(s), \quad (4.3)$$

with error $O(k2^{-k})$. At an exact integer dyadic ray $n = m2^k$, the error relative to $P(\log_2 m)$ is at most $U/(m2^k)$.

Proof. The identity in Theorem 3.2 gives $|b_n - P(\log_2 n)| \leq U/n$. Every sequence of phases has a convergent subsequence on the circle, so every cluster value belongs to the range of P . Conversely, $\log_2 \lfloor 2^k s \rfloor - k \rightarrow \log_2 s$, proving (4.3). The rounding error in the phase is $O(2^{-k})$; Proposition 3.3 turns it into $O(k2^{-k})$ in the value. A continuous real function on the circle has a compact interval as its range. Exponentiation gives the second statement of (4.2). On an exact dyadic ray there is no phase rounding, giving the stronger error bound directly. $\square$

Thus the failure of a single growth constant is organized by a continuum of phases, rather than by two isolated exceptional subsequences. The exact-period certificate in the next section will in particular show $p_- < p_+$.

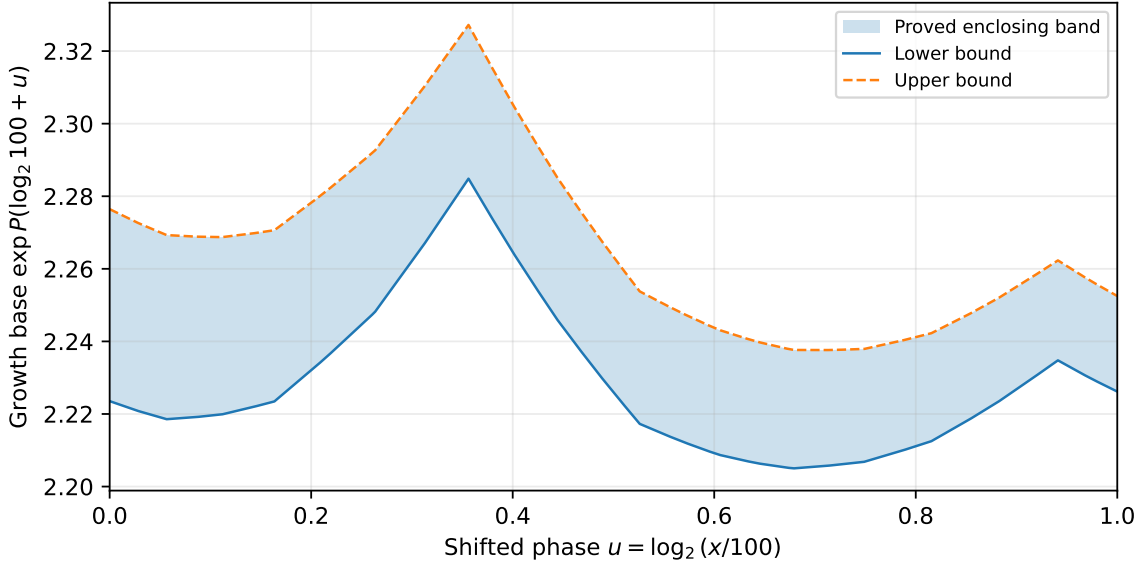


Figure 1: A proved enclosing band from the published counts $100 \leq n \leq 200$. The horizontal coordinate is shifted phase $u = \log_2(x/100)$, not the unshifted phase $\log_2 x$. Both boundaries come from (4.4); no curve is claimed to be the exact limiting profile. Plotting uses floating-point arithmetic, whereas the endpoint and period certificates use exact integers.

4.2 A certified enclosure from one dyadic-width band

Theorem 4.2 (Finite-band reconstruction). *Suppose the counts $\theta(N), \dots, \theta(2N)$ are known, where $N \geq 1$ is an integer. For every real $x \in [N, 2N]$,*

$$\frac{A(x) + L}{x} \leq P(\log_2 x) \leq \frac{A(x) + U}{x}. \quad (4.4)$$

The two envelopes are separated by at most D/N , and the interval of phases covered has length one. Furthermore,

$$\min_{N \leq n \leq 2N} \frac{a_n + L}{n} \leq p_- \leq \min_{N \leq n \leq 2N} \frac{a_n + U}{n}, \quad (4.5)$$

$$\max_{N \leq n \leq 2N} \frac{a_n + L}{n} \leq p_+ \leq \max_{N \leq n \leq 2N} \frac{a_n + U}{n}. \quad (4.6)$$

Each displayed enclosure has width at most D/N .

Proof. Equation (4.4) is (3.4) together with $Q \in [L, U]$. On a unit interval, either envelope has the form $\alpha + \beta/x$, so its maximum and minimum occur at endpoints (or it is constant). Thus extrema over the whole band can be taken at integers. Since $\log_2(2N) - \log_2 N = 1$, the band traverses every phase, which proves (4.5)–(4.6). Pointwise separation is $D/x \leq D/N$; taking maxima or minima cannot increase this bound. $\square$

The theorem gives a uniform approximation algorithm, not an efficient count enumeration algorithm. In principle one can compute all needed finite counts by exhaustive search and hence compute P uniformly, and $p_{\pm}$ as real numbers, to any prescribed accuracy. Achieving log-profile error ε requires counts through $O(\varepsilon^{-1})$ by this bound, but obtaining those counts may be very expensive.

4.3 Numerical bounds with integer verification

Applying Theorem 4.2 to $100 \leq n \leq 200$ gives the following endpoint expressions. The index selection itself can be checked without logarithms: compare $B^{1/m}$ and $C^{1/n}$ by comparing B^n and C^m .

Quantity	Certified bounding expression	Decimal approximation
lower bound for ρ_-	$(2\theta(160))^{1/160}$	2.2049986909
upper bound for ρ_-	$(21\theta(162))^{1/162}$	2.2375996902
lower bound for ρ_+	$(2\theta(128))^{1/128}$	2.2848459606
upper bound for ρ_+	$(21\theta(128))^{1/128}$	2.3272067484

In particular, the independently checked rational endpoint inequalities yield

$$2.20499 < \rho_- < 2.23760 < 2.28484 < \rho_+ < 2.32721. \quad (4.7)$$

These are enclosures, not computed exact values of the extrema. Several of the same endpoint expressions occur in Ho's analysis [5]; numerical priority is not claimed. The point of the finite-band theorem is that it certifies the entire profile and gives a systematic error bound, rather than only selected rays.

5 A finite certificate for the exact fundamental period

The previous sections prove that P is 1-periodic. A function with period one could still have period $1/2$, $1/3$, or another smaller period. Excluding this possibility is essential for the classification of *real* scale factors in (1.1).

Lemma 5.1 (Published-data phase gap). *The following inequalities hold:*

$$e^{P(0)} > \frac{57}{25} = 2.28, \quad \sup_{1/4 \leq t \leq 1/2} e^{P(t)} < \frac{9}{4} = 2.25. \quad (5.1)$$

Proof. All arithmetic needed for the proof is contained in the finite certificate

$$2\theta(128) 25^{128} > 57^{128}, \quad (5.2)$$

$$21\theta(n) 4^n < 9^n \quad (152 \leq n \leq 182). \quad (5.3)$$

These 32 strict integer inequalities are verified by `verify.py` from the published data [7]. First, 128 has phase zero, so Theorem 4.2, or (3.4) directly, gives $e^{P(0)} \geq (2\theta(128))^{1/128} > 57/25$.

Next, (5.3) says $a_n + U < n \log(9/4)$ at every integer in $[152, 182]$. Linear interpolation therefore gives $A(x) + U < x \log(9/4)$ throughout that interval. The real interval

$$[128 \cdot 2^{1/4}, 128\sqrt{2}]$$

is contained in $[152, 182]$: the lower inclusion follows from $19^4 < 2 \cdot 16^4$, and the upper from $2 \cdot 128^2 < 182^2$. For $t \in [1/4, 1/2]$, put $x = 128 \cdot 2^t$ and apply the upper envelope. Periodicity gives $P(\log_2 x) = P(7 + t) = P(t)$, proving the second inequality. The bound is uniform and strict because it holds on a compact interval. $\square$

Theorem 5.2 (Exact period group). *The set of periods of P is exactly $\mathbb{Z}$. In particular, its fundamental positive period is one.*

Proof. The phase gap makes P nonconstant. An irrational period τ , together with period one, would make P constant: the multiples of τ modulo one are dense, and continuity propagates the value at any point along that dense orbit. Thus any additional period must be rational.

If p/q in lowest terms is a period, Bézout's identity and period one show that $1/q$ is a period. Suppose $q \geq 2$. There is an integer j with

$$\frac{1}{4} \leq \frac{j}{q} \leq \frac{1}{2}. \quad (5.4)$$

For $q = 2, 3$ take $j = 1$; for $q \geq 4$ take $j = \lceil q/4 \rceil$. Period $1/q$ would imply $P(j/q) = P(0)$, contradicting (5.1). Therefore $q = 1$. Integer periods already follow from the construction, completing the proof. $\square$

The certificate proves substantially more than nonconstancy, and it does so without locating a maximum or minimum. In particular, it does *not* prove that phase zero is a global maximum. It only proves that a whole specified phase interval lies below its value, which is enough to rule out every smaller period at once.

6 Scaling rigidity and exponential gluing obstructions

6.1 Classification of all positive real scales

For $c > 0$ define, for sufficiently large integers n ,

$$T_c(n) = \log \theta(\lfloor cn \rfloor) - c \log \theta(n), \quad \Delta_c(t) = P(t + \log_2 c) - P(t). \quad (6.1)$$

Lemma 6.1 (Uniform phase formula for a scale change). *For each fixed $c > 0$,*

$$T_c(n) = cn\Delta_c(\log_2 n) + O_c(\log n). \quad (6.2)$$

If c is an integer, the remainder can be replaced by $O_c(1)$.

Proof. Let $m = \lfloor cn \rfloor$. We have $m = cn + O(1)$ and $\log_2 m = \log_2 n + \log_2 c + O_c(1/n)$. By Proposition 3.3, changing the phase by this amount changes P by $O_c(\log n/n)$. Insert this into $a_m = mP(\log_2 m) - Q(m)$, subtract $ca_n = cnP(\log_2 n) - cQ(n)$, and use boundedness of P and Q . If c is integral, there is no rounding and the remainder is exactly $cQ(n) - Q(cn)$, hence bounded. $\square$

Theorem 6.2 (Real-scale rigidity and frequent gain/loss). *For $c > 0$, $T_c(n) = o(n)$ if and only if $c = 2^j$ for an integer j . For every other c there are $\eta_c > 0$ and two sets $E_c^+, E_c^- \subseteq \mathbb{N}$ of positive lower natural density such that*

$$\theta(\lfloor cn \rfloor) \geq e^{\eta_c n} \theta(n)^c \quad (n \in E_c^+), \quad (6.3)$$

$$\theta(\lfloor cn \rfloor) \leq e^{-\eta_c n} \theta(n)^c \quad (n \in E_c^-). \quad (6.4)$$

Finite initial exceptions may be removed from these sets.

Proof. If $c = 2^j$, periodicity makes Δ_c zero, and (6.2) proves $T_c = o(n)$. Conversely, if $T_c = o(n)$, apply (6.2) along $n_k = \lfloor 2^{k+t} \rfloor$ for each fixed t . Continuity yields $\Delta_c(t) = 0$. Thus $\log_2 c$ is a period of P , and Theorem 5.2 gives $\log_2 c \in \mathbb{Z}$.

Now take an off-dyadic scale. The continuous periodic function Δ_c is not identically zero and has integral zero on a period, by translation invariance of the integral. It therefore has both positive and negative values. Continuity supplies phase intervals on which $c\Delta_c$ is bounded

below by $2\eta_c$ or above by $-2\eta_c$, for some $\eta_c > 0$. Equation (6.2) gives (6.3) and (6.4) at all sufficiently large integers with those phases.

For completeness, such phase intervals have positive lower natural density. Choose a closed mantissa interval $[u, v] \subset (1, 2)$ of positive length inside one of them. If $2^k \leq N < 2^{k+1}$, the complete dyadic blocks below 2^k contain at least

$$(v - u)(2^k - 1) - O(k)$$

integers with mantissa in $[u, v]$. Division by $N < 2^{k+1}$ gives lower density at least $(v - u)/2 > 0$. Apply this to both sign intervals. $\square$

Corollary 6.3 (Integer arities). *For an integer $m \geq 2$, a uniform subexponential comparison $\theta(mn) = \theta(n)^m \exp(o(n))$ holds exactly when m is a power of two. For $m = 2^r$ the stronger bounds*

$$2^{m-1}\theta(n)^m \leq \theta(mn) \leq 21^{m-1}\theta(n)^m \quad (6.5)$$

hold for every $n \geq 1$.

Proof. The classification is Theorem 6.2. Iterating the even recurrence r times gives factors with total exponent $1 + 2 + \dots + 2^{r-1} = m - 1$, which proves (6.5). $\square$

There is an important quantifier here: the error $o(n)$ must hold as n runs through *all* sufficiently large integers. Particular off-dyadic phases can still have zero scale difference. The theorem does not exclude such accidental equalities of profile values.

6.2 Explicit exponential failure at the split $2n + n$

Define the unequal-block gluing ratio

$$G(n) = \frac{\theta(3n)}{\theta(2n)\theta(n)}. \quad (6.6)$$

The parity recurrence gives $\log \theta(2n) - 2 \log \theta(n) \in [L, U]$, so

$$\log G(n) = 3n\Delta_3(\log_2 n) + O(1). \quad (6.7)$$

In particular, exponential loss and exponential gain for $G(n)$ each occur on a set of positive lower natural density. The following stronger explicit subsequences need only four count values.

Theorem 6.4 (Certified 2:1 gluing loss and gain). *For every integer $t \geq 0$,*

$$G(128 \cdot 2^t) < \left(\frac{49}{50}\right)^{128 \cdot 2^t}, \quad (6.8)$$

$$G(172 \cdot 2^t) > \left(\frac{21}{20}\right)^{172 \cdot 2^t}. \quad (6.9)$$

Proof. Iteration of the even recurrence gives, for $s \geq 0$,

$$2^{2^s-1}\theta(k)^{2^s} \leq \theta(k2^s) \leq 21^{2^s-1}\theta(k)^{2^s}. \quad (6.10)$$

Set

$$R_- = \frac{(21\theta(192))^2}{(2\theta(128))^3}, \quad R_+ = \frac{(2\theta(129))^4}{(21\theta(172))^3}. \quad (6.11)$$

The integer certificates checked by the accompanying program are

$$(21\theta(192))^2 50^{128} < (2\theta(128))^3 49^{128}, \quad (6.12)$$

$$(2\theta(129))^4 20^{172} > (21\theta(172))^3 21^{172}. \quad (6.13)$$

Thus $R_- < (49/50)^{128}$ and $R_+ > (21/20)^{172}$.

For $n = 128 \cdot 2^t$, write $3n = 192 \cdot 2^{t+1}$. Using the upper bound in (6.10) in the numerator and the lower bounds in the denominator of $G(n)$ gives

$$G(n) \leq \frac{4}{21} R_-^{2^t} < (49/50)^n. \quad (6.14)$$

For $n = 172 \cdot 2^t$, write $3n = 129 \cdot 2^{t+2}$. Use the lower bound for the numerator and upper bounds for both denominator factors instead. This gives

$$G(n) \geq \frac{21^2}{2} R_+^{2^t} > (21/20)^n. \quad (6.15)$$

The prefactors are respectively less than and greater than one, so no asymptotic threshold is needed: both statements hold for every $t \geq 0$. $\square$

Corollary 6.5 (No subexponential-loss universal gluing). *There is no function $\varepsilon(N) \rightarrow 0$ such that, for all sufficiently large u, v ,*

$$\theta(u+v) \geq e^{-\varepsilon(u+v)(u+v)} \theta(u) \theta(v). \quad (6.16)$$

Nor does the analogous upper bound with $e^{\varepsilon(u+v)(u+v)}$ hold uniformly for all large u, v .

Proof. Take $u = 2n, v = n$ along the loss or gain subsequence in Theorem 6.4. The proposed error exponent is $o(n)$, whereas the exhibited logarithmic discrepancy has magnitude bounded below by a positive constant times n . $\square$

The loss has a direct counting interpretation. Along the first subsequence, any map from all pairs of 3AP-free permutations of sizes $2n$ and n into 3AP-free permutations of size $3n$ must identify exponentially many pairs somewhere, if it is defined on the entire domain. In particular, a universal injective concatenation-type construction at this size ratio is impossible. This does not rule out successful constructions on a restricted family or methods that record additional output information.

7 Averaging and complete empirical limit laws

The interval of cluster values describes which limits are possible. It does not describe how often the values occur. Because the phase is logarithmic, logarithmic and ordinary sampling answer this question differently.

7.1 Logarithmic averaging has a unique limiting law

Let $H_N = \sum_{n=1}^N 1/n$ and define probability measures

$$\lambda_N = \frac{1}{H_N} \sum_{n=1}^N \frac{1}{n} \delta_{b_n}. \quad (7.1)$$

Here δ_x is the point mass at x , and convergence of probability measures means convergence against every continuous test function on a compact interval containing their supports.

Theorem 7.1 (Logarithmic empirical law). *For every continuous test function f ,*

$$\lim_{N \rightarrow \infty} \frac{1}{H_N} \sum_{n \leq N} \frac{f(b_n)}{n} = \int_0^1 f(P(t)) dt = \frac{1}{\log 2} \int_1^2 f(\Phi(s)) \frac{ds}{s}. \quad (7.2)$$

For Lipschitz f the error is $O_f(1/\log N)$. The limiting measure has support $[p_-, p_+]$. Replacing b_n by $\theta(n)^{1/n}$ pushes this law forward under the exponential map.

Proof. First take f Lipschitz. Replacing b_n by $P(\log_2 n)$ changes the unnormalized weighted sum by at most $\text{Lip}(f)U \sum n^{-2} = O_f(1)$. On $[n, n+1]$, the phase changes by $O(1/n)$, so the modulus of P gives a change $O(\log n/n)$ in its value. The difference between the resulting sum and the integral is therefore bounded by a constant times $\sum_{n \geq 2} \log n/n^2$, together with a bounded initial contribution. Consequently,

$$\sum_{n \leq N} \frac{f(b_n)}{n} = \int_1^N f(P(\log_2 x)) \frac{dx}{x} + O_f(1) \quad (7.3)$$

$$= (\log N) \int_0^1 f(P(t)) dt + O_f(1), \quad (7.4)$$

where the last step splits the logarithmic integral into complete periods and one bounded remainder. Since $H_N = \log N + O(1)$, this proves the asserted rate. Continuous test functions follow by uniform approximation by piecewise-affine, hence Lipschitz, functions on the compact range.

Every neighborhood of a value in the image of a continuous P has a preimage containing a nonempty phase interval, of positive Lebesgue measure. Thus the support is the entire image $[p_-, p_+]$. The substitution $s = 2^t$ proves the mantissa formula, and continuity of the exponential gives the final assertion. $\square$

7.2 Ordinary cutoffs retain a phase

Define the ordinary empirical measure

$$\mu_N = \frac{1}{N} \sum_{n=1}^N \delta_{b_n}. \quad (7.5)$$

For $x \in [1, 2]$ define a probability measure ν_x by its test-function integrals:

$$\begin{aligned} \nu_x(f) &= \frac{1}{x} \left(\int_1^2 f(\Phi(s)) ds + \int_1^x f(\Phi(s)) ds \right) \\ &= \frac{2}{x} \int_1^x f(\Phi(s)) ds + \frac{1}{x} \int_x^2 f(\Phi(s)) ds. \end{aligned} \quad (7.6)$$

The weights integrate to one. At the endpoints $\nu_1 = \nu_2$, as required when the cutoff phase wraps around.

Theorem 7.2 (All ordinary empirical limit laws). *For $N_k = \lfloor 2^k x \rfloor$ with $x \in [1, 2]$,*

$$\mu_{N_k} \implies \nu_x. \quad (7.7)$$

More generally, the complete set of weak subsequential limits of μ_N is $\{\nu_x : 1 \leq x \leq 2\}$. The map $x \mapsto \nu_x$ is weakly continuous and all its measures have support $[p_-, p_+]$.

Proof. For Lipschitz f , replacing b_n by $P(\log_2 n)$ changes its ordinary average by $O_f(H_N/N)$, which tends to zero. Continuous f again follow by uniform approximation. Put $u_n = n/2^k$. Since P is 1-periodic, $P(\log_2 n) = P(\log_2 u_n)$. Riemann sums on $[\varepsilon, x]$, followed by $\varepsilon \downarrow 0$, give

$$\lim_{k \rightarrow \infty} \mu_{\lfloor 2^k x \rfloor}(f) = \frac{1}{x} \int_0^x f(P(\log_2 u)) du. \quad (7.8)$$

The integrand is bounded and continuous away from zero, so the omitted interval has arbitrarily small contribution.

Let $C_f = \int_1^2 f(\Phi(s)) ds$. Splitting $(0, 1]$ into dyadic intervals and substituting $u = 2^{-j}s$ shows

$$\int_0^1 f(P(\log_2 u)) du = \sum_{j \geq 1} 2^{-j} C_f = C_f. \quad (7.9)$$

Thus (7.8) equals (7.6).

The same Riemann-sum argument holds for a sequence of upper endpoints $x_k \rightarrow x$, by boundedness of the integrand and the small length of the interval between x_k and x . Every sequence of cutoffs has a subsequence whose mantissas $N/2^{\lfloor \log_2 N \rfloor}$ converge in $[1, 2]$. This proves that no other weak limit is possible, while the fixed-mantissa cutoffs realize every ν_x . Formula (7.6) proves weak continuity. Its mantissa density is at least $1/2$ throughout $[1, 2]$, so the support is the full image of Φ , by the same neighborhood argument used for the logarithmic law. $\square$

Theorem 7.3 (Ordinary averages do not wash out the oscillation). *Neither of the ordinary Cesàro averages*

$$\frac{1}{N} \sum_{n \leq N} \frac{\log \theta(n)}{n}, \quad \frac{1}{N} \sum_{n \leq N} \theta(n)^{1/n} \quad (7.10)$$

converges. In particular, neither corresponding ordinary empirical measure has a single weak limit.

Proof. For the first average its cutoff-dependent limit is

$$M(x) = \frac{C + \int_1^x \Phi(s) ds}{x}, \quad C = \int_1^2 \Phi(s) ds. \quad (7.11)$$

Continuity of Φ makes M continuously differentiable, with

$$M'(x) = \frac{\Phi(x) - M(x)}{x}. \quad (7.12)$$

If M were constant, (7.12) would force Φ to be constant, contradicting the phase gap. Thus two cutoff phases give distinct limiting means. For the second average replace Φ throughout by e^Φ , which is likewise continuous and nonconstant. Nonconvergence of these continuous test-function integrals precludes weak convergence of the empirical measures. $\square$

Remark 7.4. Full support is not the same as existence of a probability density. The pushforward laws could in principle have atoms at values corresponding to positive-measure level sets of P . Nothing in the arguments above proves absolute continuity, absence of atoms, or a particular shape of a histogram.

8 Polynomial and geometric sampling

8.1 Multiplicatively fine sampling preserves the spectrum

Proposition 8.1. *Let n_k be a strictly increasing unbounded sequence of positive integers with $n_{k+1}/n_k \rightarrow 1$. Then*

$$\text{Clust}(b_{n_k}) = [p_-, p_+], \quad \text{Clust}(\theta(n_k)^{1/n_k}) = [\rho_-, \rho_+]. \quad (8.1)$$

In particular this holds along every eventually positive arithmetic progression with positive common difference and every eventually positive, nonconstant integer-valued polynomial sampled at positive integers.

Proof. Fix $s \in [1, 2]$ and let $n_{k(j)}$ be the first sample not below $2^j s$. Its predecessor is below $2^j s$, so

$$1 \leq \frac{n_{k(j)}}{2^j s} \leq \frac{n_{k(j)}}{n_{k(j)-1}} \rightarrow 1.$$

Consequently its phases converge to $\log_2 s$ modulo one, and the profile law gives limit $\Phi(s)$. These values fill the interval; no new values are possible because the sampled sequence is a subsequence of the original. A polynomial of positive degree and positive leading coefficient is strictly increasing eventually and has successive ratios tending to one. Arithmetic progressions are the degree-one case. Removing a finite initial segment changes none of the assertions. $\square$

This assertion concerns the cluster set, not the frequency distribution under polynomial sampling. It already shows that the continuum of growth rates is not confined to specially chosen powers of two.

8.2 Geometric sampling is governed by rotation

Theorem 8.2 (Geometric sampling classification). *Let $c > 0$, $b > 1$, $n_k = \lfloor cb^k \rfloor$, and put $\alpha = \log_2 b$, $\beta = \log_2 c$. Ignore any initial zero terms.*

(i) *If $\alpha = p/q$ is rational in lowest terms, the cluster values of b_{n_k} are*

$$\{P(\beta + jp/q) : 0 \leq j < q\}, \quad (8.2)$$

and the empirical law is the average of the q corresponding point masses. Some of the displayed values may coincide.

(ii) *If α is irrational, the cluster set is $[p_-, p_+]$, and*

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=1}^K f(b_{n_k}) = \int_0^1 f(P(t)) dt \quad (8.3)$$

for every continuous f .

Both statements transfer to $\theta(n_k)^{1/n_k}$ by exponentiation.

Proof. Rounding gives $\log_2 n_k = \beta + k\alpha + O(b^{-k})$. The modulus of continuity and bounded correction imply

$$b_{n_k} = P(\beta + k\alpha) + O(kb^{-k}). \quad (8.4)$$

In the rational case the phase sequence is periodic with period q , and each residue class of k has frequency $1/q$.

For irrational α , each nonzero integer h satisfies

$$\frac{1}{K} \sum_{k=1}^K e^{2\pi i h(\beta + k\alpha)} \longrightarrow 0,$$

by the finite geometric-sum formula, because $e^{2\pi i h\alpha} \neq 1$. Approximation of continuous periodic functions by trigonometric polynomials gives uniform distribution of these phases. Apply it to $f \circ P$. The orbit is dense, so continuity also gives the full cluster interval. The vanishing rounding error changes neither the cluster sets nor the limiting averages. $\square$

For example, sampling $n = 3^k$ produces the full growth interval and the logarithmic limiting law, because $\log_2 3$ is irrational. In contrast, sampling $n = 2^k$ converges to the single phase-zero growth value. Exact fundamental period one does not mean that different phases always have different values: a continuous real periodic function cannot be injective on its circle.

9 Local ratios and the remaining monotonicity problem

The profile theorem should not be confused with a proof that $\theta(n)$ increases at every step. Ho explicitly distinguishes that open question from the root-growth question [5]. The following exact identity shows where the difficulty resides.

Proposition 9.1 (Binary descent for adjacent logarithmic ratios). *Let $d_n = a_{n+1} - a_n$. Then $d_1 = L$, and for $n \geq 2$,*

$$d_n = d_{\lfloor n/2 \rfloor} + r_{n+1} - r_n. \quad (9.1)$$

Consequently,

$$L - D \lfloor \log_2 n \rfloor \leq d_n \leq L + D \lfloor \log_2 n \rfloor, \quad (9.2)$$

and, with $\gamma = \log_2(21/2)$,

$$2n^{-\gamma} \leq \frac{\theta(n+1)}{\theta(n)} \leq 2n^\gamma. \quad (9.3)$$

The elementary deletion bound improves the upper side to $n+1$.

Proof. For $n = 2m$, subtract the recurrences for a_{2m+1} and a_{2m} ; for $n = 2m+1$, subtract those for a_{2m+2} and a_{2m+1} . Both subtractions produce (9.1). Iteration reaches index one after exactly $\lfloor \log_2 n \rfloor$ steps. Each toll difference is in $[-D, D]$, proving (9.2); exponentiation gives (9.3). For the deletion bound, remove the entry $n+1$ from a 3AP-free permutation of $[n+1]$. The remaining ordering avoids all progressions in $[n]$. Together with its original insertion position, this is an injective encoding into at most $(n+1)\theta(n)$ possibilities. $\square$

The lower bound in (9.3) is positive but falls below one; it therefore does not solve monotonicity. Proving $d_n \geq 0$ would require more information about the correlated toll differences along binary descent than the independent bounds $r_n \in [L, U]$ supply.

The nondegenerate growth interval also prevents eventual log-convexity or eventual log-concavity. Either property would make adjacent logarithmic ratios eventually monotone. The exponential bounds on a_n would then force their Cesàro means a_n/n to converge, contradicting Theorem 4.1. This is a limitation of possible regularity arguments, not a substitute for the unresolved monotonicity question.

Remark 9.2 (Adjacent ratios have no limit [write]). The same Cesàro argument shows that $\theta(n+1)/\theta(n)$ has no limit in $[0, \infty]$. A finite positive limit λ would give $d_n \rightarrow \log \lambda$ and hence $b_n \rightarrow \log \lambda$, contradicting Theorem 4.1 with $p_- < p_+$ (or Ho's theorem [5]); the limits 0 and

∞ would force $b_n \rightarrow -\infty$ and $b_n \rightarrow +\infty$ respectively, contradicting (2.6). The special case that the ratio does not tend to 2 is machine-checked in PROVEIT (Section 11); the general statement is not formalized. The upper side of Proposition 9.1 is also improved in Lean, to $\theta(n+1) \leq \lfloor (n+3)/2 \rfloor \theta(n)$.

10 Reproducible verification and a formalization route

10.1 What the package checks

The program `verify.py` uses only the Python standard library. Its executed record is `verification.json`. It performs the following logically distinct tasks.

1. It checks all 199 lower and upper splitting inequalities in the supplied range $2 \leq n \leq 200$. These are consistency checks on a finite table, not a proof of the infinite recurrence.
2. It checks the 32 phase-gap inequalities, the two interval-containment inequalities, the two explicit gluing certificates, and two previously used Ho separation certificates, with exact integer arithmetic.
3. It chooses the extremizing indices in each band $[32, 64]$, $[64, 128]$, and $[100, 200]$ by exact power comparisons. It encloses each selected root between decimal rationals by integer bisection. Decimal logarithms are used only to print additional approximations.
4. It independently recomputes $\theta(n)$ for every $0 \leq n \leq 32$ using a subset-state recurrence, and cross-checks $0 \leq n \leq 8$ by direct permutation enumeration.

All these executed checks passed. No claim is made that the independent Python enumeration was run for $33 \leq n \leq 200$, or that a Python assertion is a proof checked by a theorem prover.

[write] In the collection the program is shipped as `code/verify.py` and its executed record as `data/verification.json`; the counts are `data/counts.txt`. The program locates its data relative to its own directory, so it must be run from a copy that restores the delivered flat layout (program beside `data/`); the report's README gives the commands. A rerun of `verify.py -max-n 32` on such a copy, made when the report was written, passed and reproduced the shipped record except for its timing fields.

10.2 Why the independent enumeration is correct

For a fixed n , represent the set of values already placed in a prefix by $S \subseteq [n]$. A candidate next value $y \notin S$ is allowed if, for every $d \geq 1$ with $y - d, y + d \in [n]$, either both endpoints belong to S or neither does. Equivalently,

$$\mathbf{1}_S(y - d) = \mathbf{1}_S(y + d). \quad (10.1)$$

Let $C(S)$ be the number of continuations generated by this rule, with $C([n]) = 1$ and

$$C(S) = \sum_{\substack{y \notin S \\ y \text{ allowed}}} C(S \cup \{y\}). \quad (10.2)$$

Then $C(\emptyset) = \theta(n)$.

To prove this, consider a complete permutation and a prospective middle value y of an arithmetic progression. At the moment y is placed, that progression appears as a subsequence if and only if exactly one of its two endpoints has already appeared. Thus (10.1) excludes

exactly the forbidden full permutations. Every full permutation corresponds to exactly one chain of subsets and choices, so (10.2) counts each permitted permutation once. The condition depends only on the used set, which justifies memoization; no unrecorded ordering information is needed for this rule. This is an independently written implementation of the standard subset-state idea, not a claim of a new enumeration algorithm [4].

10.3 Certificate design for a proof assistant

A practical formalization can be layered rather than beginning with the analytic limit.

First prove the finite permutation definition, the parity injection, and soundness of (10.2). Next expose Sharma’s estimate as an explicit imported theorem or an explicitly named hypothesis until its full combinatorial proof has been formalized. Define a_n and r_n , prove the piecewise-affine interpolation lemma, and construct F by uniform convergence of the geometric tail. This supplies the periodic profile without any numerical experiments.

The period and gluing proofs then reduce to a small finite certificate over natural-number powers, plus the already proved interpolation envelope. A verified certificate evaluator should distinguish “the table entry is the count” from “the inequality involving that entry is true.” Finally, formalize the density and integral arguments on compact intervals. This separation prevents a conditional analytic theorem from being reported as an unconditional theorem about the combinatorial sequence.

The present package contains no Lean file asserting these results. The route just described is an implementation plan, not an executed formalization.

[write] Part of the first layer already exists in PROVEIt, and the second layer’s “imported theorem” is not needed there: the permutation definition, the parity construction and both parity recurrences, the count identification between the three papers’ definitions, and Sharma’s Theorem 2.8 itself are proved in Lean (Section 11). The soundness of the subset-state recurrence (10.2), the analytic profile, and every later layer are not formalized. The existing finite-count evaluator stops at $n = 20$ and runs through `native_decide` on a native backend, which is itself a trust boundary; the certificates above need n between 100 and 200.

11 Formal status in ProveIt [write]

This section was added when the report entered the collection. It records what PROVEIt’s Lean development of the three source papers [1, 2, 3] already proves about the objects of this article, and what it does not. (The Lean namespace `Sharma2012` formalizes the paper cited here as [2], 2009.) The Lean files are under `Combinatorics/Ramsey/Lean/`; the paths below are relative to that directory, the namespaces begin with `LeanProofs.`, and line numbers are those of the current tree, which agrees with the manuscript’s pinned revision in this directory. No Lean build and no axiom audit were run for this report.

11.1 Declarations that prove inputs of this article

- `Sharma2012.theorem_2_8_holds` (`Sharma2012/FinalConsequences.lean:23`), proving `Sharma2012.theorem_2_8` (`Sharma2012/Statements.lean:272`):
 $\theta(n) \leq 21 \theta(\lfloor (n+1)/2 \rfloor) \theta(\lfloor n/2 \rfloor)$ for $n \geq 3$. This is (2.2), the one theorem the manuscript imports. The case $n = 2$ is not part of the Lean statement.
- `DavisEntringerGrahamSimmons1977.even_count_recurrence_holds` and `odd_count_recurrence_holds`

(DavisEntringerGrahamSimmons1977/CountingRecurrences.lean, lines 252 and 485): $2M(k)^2 \leq M(2k)$ and $2M(k+1)M(k) \leq M(2k+1)$ for $k \geq 1$. Together they are (2.1) for every $n \geq 2$. The same inequalities are restated for Sharma's θ as Sharma2012.inequality_1_holds and inequality_2_holds (Sharma2012/CountingConsequences.lean, lines 19 and 24) and for LeSaulnier–Vijay's M as LeSaulnierVijay2011.M_even_recurrence_holds and M_odd_recurrence_holds (LeSaulnierVijay2011/Proofs.lean, lines 94 and 100).

- Sharma2012.theta_eq_davis_count (Sharma2012/AsymptoticConsequences.lean:22): Sharma's θ and the Davis–Entringer–Graham–Simmons M are the same count.
- Sharma2012.theorem_1_1_holds and DavisEntringerGrahamSimmons1977.fact_1_holds: $2^{n-1} \leq \theta(n)$ for $n \geq 1$, the lower half of (2.6). The upper half, $\theta(n) \leq 21^{n-1}$, is not stated in Lean; Sharma's theorem_2_9_holds proves the different bound $\theta(n) \leq (27/10)^n/21$ for $n \geq 11$.
- DavisEntringerGrahamSimmons1977.table_1_holds: the values $\theta(1), \dots, \theta(20)$, which agree with the first twenty entries of data/counts.txt. They rest on RamseyPaperCommon.reflectedM_values_through_twenty (RamseyPaperCommon/CountCertificates.lean:17), proved by native_decide through an implemented_by native backend (RamseyPaperCommon/ThreeFreeCounting.lean, lines 359–374). These values therefore carry that explicit trust boundary. Whether the declarations above depend on it was not audited here; the import closure of Sharma2012/FinalConsequences.lean contains it.

11.2 Declarations bearing on the article's discussion

- DavisEntringerGrahamSimmons1977.insertion_count_upper_bound_holds (DavisEntringerGrahamSimmons1977/Proofs.lean:4824): $\theta(n+1) \leq \lfloor (n+3)/2 \rfloor \theta(n)$ for $n \geq 1$. This is stronger than the deletion bound $(n+1)\theta(n)$ at the end of Proposition 9.1.
- LeSaulnierVijay2011.M_ratio_does_not_tend_to_two_holds (LeSaulnierVijay2011/Proofs.lean:231): $\theta(n+1)/\theta(n)$ does not tend to 2. This is the negative answer to the question Sharma records as Sharma2012.introduction_ratio_question. Remark 9.2 below derives the stronger unformalized statement that the ratio has no limit at all.
- LeSaulnierVijay2011.theorem_1_holds: $\theta(n) \geq \frac{1}{2} 2132^{n/10}$ for $n \geq 8$, and Sharma's theorem_2_10_holds: $\limsup \theta(n)^{1/n} \leq 27/10$. These are the formal exponential bounds. The enclosure (4.7) is narrower but rests on published counts that PROVEIt does not certify.
- Sharma2012.open_problem_1 (monotonicity of θ) and open_problem_2 (convergence of $\theta(n)^{1/n}$) (Sharma2012/Statements.lean, lines 312 and 316) have no proof declarations, by design (Sharma2012/Proofs.lean, lines 6–9). The first is still open, as Section 9 says. The second is answered negatively by Ho's theorem [5], and again by Theorems 4.1 and 5.2; neither answer is formalized, and the Lean docstring correctly describes the question as open in Sharma's paper. open_problem_3, whether $\theta(n+1) < 3\theta(n)$ always, is untouched by this article.

11.3 What is not formalized

None of the article’s own results is formalized: the bounded-toll profile (Theorem 3.2), its modulus, the spectrum and band theorems, the exact period (Theorem 5.2), the real-scale classification (Theorem 6.2), the explicit gluing certificates, the empirical and sampling laws, the binary-descent identity, and the soundness of the subset-state enumeration. The integer certificates are checked by a Python program only, from published counts that PROVEIT does not certify. Placement of this report beside the Lean development gives none of these statements formal status.

12 Further research questions and topics

The results narrow several questions without resolving all of the fine structure. The following proposed program separates concrete next steps from more speculative extensions.

1. **Extremizing phases and sharp constants.** Determine the exact values or substantially narrower enclosures for $p_{\pm}$, and characterize their maximizing and minimizing phase sets. The certificate used here does not show that $P(0)$ is a global maximum, nor that either extremum has a unique phase. Larger count bands immediately produce better certified enclosures, while a structural argument could be far more informative than additional enumeration.
2. **Regularity beyond the general bound.** Is P Lipschitz? Is it differentiable at typical phases, at any dyadic phase, or on a set of full measure? Is the logarithmic loss in (3.11) actually necessary? These questions require information about cancellations between toll increments; boundedness alone does not settle them.
3. **The distributions of growth values.** Determine whether the logarithmic pushforward law has atoms, a density, or a singular component. Study the dependence of the ordinary laws ν_x on x , including when two distinct cutoff phases produce the same law. The formula (7.6) is exact, but it does not determine the regularity of the pushforward without further knowledge of level sets of P .
4. **The bounded multiplicative correction.** Understand $Q(n)$ and the tolls r_n along dyadic and other arithmetic subsequences. A limiting correction would strengthen a phase-dependent exponential envelope to an asymptotic equivalence on that subsequence. Nothing proved here implies such convergence, and a nontrivial limiting law for the corrections may be more realistic than a constant prefactor.
5. **Monotonicity and adjacent ratios.** Decide whether $\theta(n+1) \geq \theta(n)$ always holds. The binary-descent identity (9.1) offers a precise target: control a sum of correlated toll differences, not merely each difference separately. A combinatorial injection would resolve monotonicity but must accommodate the unequal-size exponential obstructions proved in this paper. [write] This is Sharma’s open problem (1), recorded without proof in PROVEIT as `Sharma2012.open_problem_1`.
6. **Sharper interleaving bounds and effective computation.** Can the uniform factor 21 be reduced, perhaps beyond a finite threshold, or replaced by a phase-sensitive bound? Either improvement would sharpen all the enclosures automatically. Separately, develop an independently checkable enumeration pipeline for larger count bands with explicit memory, time, and certificate-size accounting.

7. **Machine-checked end-to-end certificates.** Formalize the complete implication from a definition of θ and a verified finite-count evaluator to the phase gap, exact period, and gluing inequalities. The analytic construction is reusable for other balanced recurrences. The main proof-engineering challenge is avoiding hidden large count assumptions while keeping the certificate small. [write] In PROVEIT the definition of θ , both splitting recurrences (including Sharma’s upper one) and the values through $n = 20$ are already formalized (Section 11). What remains is the verified evaluator for the counts the certificates use ($100 \leq n \leq 200$) without a native trust boundary, the analytic profile, and the certificate implications themselves.
8. **Other avoidance classes and generating functions.** Search for analogous bounded-toll recurrences in longer-progression avoidance or other permutation classes. Ask whether the resulting profile can have a smaller fundamental period or whether an exact scale group can again be read from finite data. For $\sum \theta(n)z^n$, investigate what the nonconstant exponential profile implies about analytic continuation or algebraic/differential equations. No non-D-finiteness or natural-boundary claim is proved here.

13 Conclusion

A nonexistent growth constant need not signal unstructured behavior. For 3AP-free permutations, the balanced combinatorial recurrence yields a unique continuous logarithmic phase profile with a bounded correction. Published finite data do more than distinguish two limits: they certify the profile’s exact fundamental period. That finite certificate determines the entire real scaling-symmetry group and forces exponential gain and loss at every off-dyadic scale.

The explicit 2:1 gluing inequalities show a concrete consequence for combinatorial constructions, while the empirical laws explain which forms of averaging preserve or erase the cutoff phase. These results leave the fine regularity, precise extrema, bounded corrections, and monotonicity of the counting sequence as well-defined further problems rather than concealing them inside an unsupported asymptotic ansatz.

14 Provenance and place in the collection [write]

14.1 Source

This report was built from **one** manuscript: manuscript 06 of the fortieth intake batch of PROVEIT’s incoming reports, delivered as the archive `ProveIt_Dyadic_Scaling_Rigidity.zip` (arrival commit `017e6c0d8`), dated September 28, 2026, and pinned to PROVEIT revision `74f7f5bdb` (full hash in Section 1.3 and in the bibliography). It was placed as a new single-source report in commit `afb2d1227`. It contributed everything in this article except the text marked [write]. No merge was involved, so there were no choices between sources.

When the report was written, every label received the prefix `dsr::`; the text marked [write] was added (the collection note, the correction in Remark 1.1, the notation table of Section 1.5, Remarks 2.4 and 9.2, Section 11, short notes in Sections 1, 10 and 12 and in the appendix, and this section). No statement, proof, number, limitation or non-claim of the manuscript was removed or changed, and no symbol was renamed. The delivered program, data, figure and audit are shipped byte-identical; their shipped paths are listed in the report’s README, and the delivered audit `SOURCE_AND_PROOF_AUDIT.md` still says that the Lean catalogue was not checked for proofs, which Remark 1.1 corrects. The delivered PDF and checksum ledger are not shipped; this article’s PDF is a build of this text.

14.2 Neighbouring reports

No other report in PROVEIT’s collections treats A003407 or 3AP-free permutations. Within `oeis-sequence-asymptotics`, the closest contrast is `a158415-growth-constant`, where a growth constant $\lim a(n)^{1/n}$ is proved to exist; here none exists, and the periodic profile P takes its place. The formal neighbour is the Lean development under `Combinatorics/Ramsey/Lean/` described in Section 11.

A Minimal arithmetic certificate and reproducibility

The core exact-period certificate can be expressed as the following short fragment, where `theta` is the supplied dictionary of published counts. No logarithms or floating-point comparisons occur.

```
assert 2 * theta[128] * 25**128 > 57**128
assert 19**4 < 2 * 16**4
assert 2 * 128**2 < 182**2
for n in range(152, 183):
    assert 21 * theta[n] * 4**n < 9**n

assert (21*theta[192])**2 * 50**128 < \
    (2*theta[128])**3 * 49**128
assert (2*theta[129])**4 * 20**172 > \
    (21*theta[172])**3 * 21**172
```

The complete program validates the input structure, recomputes small counts, checks additional consistency and endpoint certificates, and writes a machine-readable record. To reproduce the executed checks and rebuild the article, run from the package directory:

```
python verify.py --max-n 32
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The existing figure is included in the package. Regeneration with `python plot_profile.py` additionally requires NumPy and Matplotlib; neither is needed for the exact arithmetic checks. The higher-precision printed decimals are supplementary to the strict rational enclosures. [write] These commands assume the delivered flat layout. In the collection the scripts are in `code/`, so run them on a copy as the report’s README describes; running `code/verify.py` in place fails to find its data.

References

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- [2] A. Sharma, *Enumerating permutations that avoid three term arithmetic progressions*, Electronic Journal of Combinatorics **16**(1) (2009), #R63. <https://doi.org/10.37236/152>. Theorem 2.8 is the upper splitting estimate used here.
- [3] T. D. LeSaulnier and S. Vijay, *On permutations avoiding arithmetic progressions*, Discrete Mathematics **311** (2011), 205–207. <https://doi.org/10.1016/j.disc.2010.10.006>.

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<https://github.com/VladimirReshetnikov/ProveIt>. Inspected revision 74f7f5bdba1aa728b340509701a0fa4e45e8dbf3; Ramsey statement catalogues and repository overview. Targeted inspection only; no full repository build or exhaustive novelty audit.